Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/15/2005
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Aerodynamics and Chemical Kinetics of Premixed Flames at High Pressures		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	note: renewed 4 yrs		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04AA09A		
Performance Goal No.:			
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Task Description:	The program consists of several parts, each of which consists of both experimental measurements and direct numerical simulation of microgravity flames: 1) Determination of the laminar flame speeds of gaseous fuels; 2) Determination of the corresponding Markstein lengths for these flames; 3) Determination of the extinction limits; 4) Investigation of Landau-Darrieus and diffusional-thermal flame surface instabilities; and 5) Recompilation of existing chemical kinetic mechanisms by using the high-pressure flame speed data as optimization points. The purpose of this research is to develop a fundamental understanding of premixed gas flame propagation, extinction and stability in high-pressure environments, where microgravity experiments will permit measurements undisturbed by convective effects. The chemical reaction mechanisms, which govern the various flame processes, are inherently nonlinear, and can cause non-monotonic variations of flame responses to increased pressure. Most practical combustion systems, especially internal combustion engines, operate at elevated pressures; while the fundamental information on flame chemistry were		

	obtained at substantially lower pressures. Contributions to the fundamental understanding of flame processes at elevated pressures will have direct relevance to advances in the development of efficient and low-emission internal combustion engines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Contributions to the fundamental understanding of flame processes at elevated pressures will have direct relevance to advances in the development of efficient and low-emission internal combustion engines.
Task Progress:	Markstein lengths and stretch-free laminar flame speeds were extracted for smooth acetylene, ethylene, ethane, propylene, propane, and dimethyl ether (DME) flames, nominally up to 10 atmospheres. These archival data are of relevance in the design of high-pressure combustion processes, and as target points for optimization of detailed kinetic mechanisms. Except for the results on DME, the experimental and numerical results show reasonable agreement for lean flames. The discrepancies for rich flames are larger, and in particular the rich ethylene chemistry was not well captured by the kinetic mechanism. Further, the pressure coefficients for the key reactions in the mechanism were not properly captured, yielding relative discrepancies between the different pressures for a given fuel and equivalence ratio. The scatter in the results for DME is subject to further research, both numerically and experimentally. We have in addition conducted an experimental and theoretical investigation of the phenomenon of spontaneous cell formation on expanding acetylene, ethylene, hydrogen, propane, and hydrogen/propane spherical flames at elevated pressures. Critical conditions for the onset of instability were measured, and results were generally expressed in terms of a Peclet number, defined as the ratio of flame radius to the flame thickness. In addition, an asymptotic theory on the stability of spherically expanding flames was developed for a mixture consisting of two scarce fuels with distinct Lewis numbers. Predicted values of Peclet numbers for both pure fuels and fuel mixtures were shown to agree reasonably well with the measured values. For the pure fuels, studying near-unity Lewis number flames of acetylene and ethylene isolated the effects of the hydrodynamic from the Lewis number dependent diffusional instabilities. It was shown that the flame thickness was the strongest governing parameter in the absence of significant diffusional-thermal instabilities. The study on pure fuels further yielded interesting results on the critical stretch rate for the onset of instabilities, and as such, non-equidiffusion was clearly shown through skew in these raw data. This result has not been observed for laminar flames before, but is often reported in turbulent flame studies as it is very relevant to the understanding of flames in the laminar flamelet regime of turbulent combustion modeling. Finally, all the instability results were presented in a fully non-dimensional space, namely a Markstein-Karlovitz space. The results for both pure fuels and fuel mixtures fell on almost the same curve, indicating that the growth rate of instabilities was captured for the substantial variation in operating conditions, and only a function of the flame speed. This result may be somewhat fortuitous in that it was shown through a parametric study that the growth rate of the instabilities was not a strong function of the expansion ratio for the flames in the present investigation. Sivashinsky's theory, however, predicts that the growth rate is a function of both the expansion ratio and the flame speed. Given the shortcoming of the Wang-mechanism for the simplest reactions, new studies on, for example hydrogen with carbon monoxide, could possibly provide valuable target points for an improved optimization study. It is also clear that further investigation on the large discrepancies for dimethyl ether is needed, both experimentally and numerically. A major goal of our study is to use the knowledge gained to test outwardly propagating flames in a turbulent flow field. It has been shown to some extent that there is a strong influence of preferential diffusion/stretch interactions on turbulent flame front propagation and that the Markstein number determined from quasi-steady laminar flames is an adequate parameter to quantify the influence of integral strain rate on the space-averaged laminar burning velocity. Mapping of cell size and evolution is a target for future research, both to elucidate the nature behind cellular instabilities and because of its obvious relevance to turbulent combustion. Palm-Leis & Strehlow noted a tendency for turbulence to be more effective in enhancing flame speeds when the flame was inherently unstable, and given our results from laminar flames, we expect this to be the case, actually, it is rather obvious. Further, Istratov & Librovich have suggested a sequential development of cellular structures in a spherical laminar flame that then becomes unstable, leading to turbulence generation and, ultimately, detonation, and explorations along this path would also be of great value to the combustion community at large.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
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